MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROPOLYMER-INSULATED, TFE-GLASS-FEP, MEDIUM WEIGHT, SILVER-COATED COPPER CONDUCTOR, 600-VOLT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of Specification MIL-W-22759.

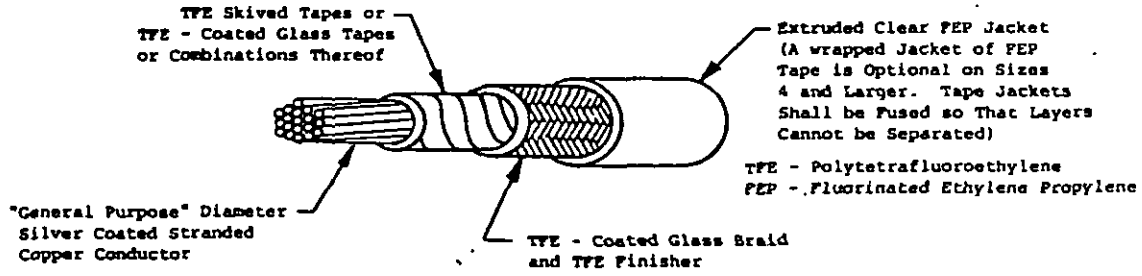


TABLE I. CONSTRUCTION DETAILS

Part No. 1/	Wire size	Stranding (Number of strands X AWG gage of strands)	Diameter of stranded conductor (inches)		Finished wire			
			(min)	(max)	Resistance at 20°C(68°F) (ohms/1000 ft) (max)	Thickness of FEP jacket (inches)	Diameter (inches)	Weight (lbs/1000 ft) (max)
M22759/4-22-*	22	19 X 34	.029	.032	15.1	.009 ±.002	.074 ±.003	5.90
M22759/4-20-*	20	19 X 32	.037	.040	9.19	.009 ±.002	.082 ±.003	7.90
M22759/4-18-*	18	19 X 30	.046	.050	5.79	.010 ±.002	.095 ±.003	11.0
M22759/4-16-*	16	19 X 29	.052	.057	4.52	.010 ±.002	.103 ±.004	(B) 13.6
M22759/4-14-*	14	19 X 27	.065	.072	2.88	.010 ±.002	.116 ±.004	18.5
M22759/4-12-*	12	37 X 28	.084	.089	1.90	.010 ±.002	.133 ±.004	26.6
M22759/4-10-*	10	37 X 26	.106	.112	1.19	.010 ±.002	.164 ±.006	41.5
M22759/4-8-*	8	133 X 29	.158	.169	.658	.010 ±.002	.235 ±.007	(B) 77.4
M22759/4-6-*	6	133 X 27	.198	.213	.418	.010 ±.002	.282 ±.010	115.
M22759/4-4-*	4	133 X 25	.250	.268	.264	.012 ±.003	.351 ±.015	(B) 184.
M22759/4-2-*	2	665 X 30	.320	.340	.170	.012 ±.003	.430 ±.015	(B) 281.
M22759/4-1-*	1	817 X 30	.360	.380	.139	.014 ±.004	.480 ±.015	(B) 358.
M22759/4-01-*	0	1045 X 30	(B) .395	.425	.108	.014 ±.004	.525 ±.025	(B) 436.
M22759/4-02-*	00	1330 X 30	(B) .440	.475	.085	.014 ±.004	.585 ±.025	(B) 554.

1/. PART NO.: The asterisks in the part number column, Tables I through III, shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white - M22759/4-20-9; white with orange stripe - M22759/4-20-93.

(B) denotes changes

FSC 6145

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TABLE II. PERFORMANCE DETAILS

Part No.	Abrasion resistance (Procedure I)					Bend testing				
	Resistance (inches of tape) (min)		Weight support bracket	Weight (lbs)	Tension load (lbs)	Mandrel diameter (inches) (±3%)			Test load (lbs) (±3%)	
	Initial	After immersion 1/				Life cycle (oven & bend tests) 2/	Cold bend test	Wrap test	Life cycle (oven & bend tests) 2/	Cold bend test
M22759/4-22-*	18	13	A	1.0	1.0	.375	.50	.250	.75	2.0
M22759/4-20-*	18	13	A	1.0	1.0	.375	.50	.250	.75	2.0
M22759/4-18-*	24	17	A	1.0	1.0	.50	.75	.375	1.0	2.0
M22759/4-16-*	30	21	A	1.0	2.0	.50	.75	.375	1.0	3.0
M22759/4-14-*	12	8	B	3.0	2.0	.50	1.00	.50	1.0	3.0
M22759/4-12-*	12	8	B	3.0	2.0	.75	2.00	.50	3.0	3.0
M22759/4-10-*	18	13	B	3.0	2.0	1.00	3.00	.75	3.0	5.0
M22759/4-8-*	24	17	B	3.0	2.0	1.50	3.00	1.00	3.0	5.0
M22759/4-6-*	30	21	C	3.0	2.0	2.00	4.00	1.25	6.0	10.0
M22759/4-4-*	36	25	C	4.25	2.0	3.00	6.00	1.50	6.0	10.0
M22759/4-2-*	40	28	C	4.25	2.0	4.00	6.00	2.00	6.0	15.0
M22759/4-1-*	40	28	C	4.25	2.0	6.50	10.00	2.50	6.0	15.0
M22759/4-01-*	48	34	C	4.25	2.0	10.00	10.00	3.00	10.0	20.0
M22759/4-02-*	48	34	C	4.25	2.0	10.00	18.00	4.00	10.0	25.0

1/ Qualification test only.

2/ Also for bend tests after immersion.

ADDITIONAL REQUIREMENTS

TEMPERATURE RATING: 200°C (392°F) max conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level
SPARK TEST OF PRIMARY INSULATION: Not required
IMPULSE DIELECTRIC TEST: 8.0 kilovolts (peak), 100% test
INSULATION RESISTANCE: 5000 for 1000 ft (min)
WRAP TEST: Mandrel wrap test required; no cracking
Dielectric test after mandrel wrap, 2500 volts (rms), 60Hz
BLOCKING: 200 ±2°C (392 ±3.6°F)
SHRINKAGE: 0.125 inch max at 200 ±2°C (392 ±3.6°F)
WICKING:
Sizes 22 through 12 - Procedure II; 2.0% (max) weight increase, 0.750 inch (max) dye travel
Sizes 10 through 00 - No requirement
LOW TEMPERATURE (COLD BEND):
Bend temperature: -65 ±2°C (-85 ±3.6°F)
Dielectric test, 2500 volts (rms), 60Hz
THERMAL SHOCK: Oven temperature, 200 ±2°C (392 ±3.6°F)
Max change in measurement Sizes 22 through 12: 0.060 inch
Sizes 10 through 8: 0.100 inch
Size 6 through 00: 0.125 inch
FLAMMABILITY: Post-flame electric test not required
LIFE CYCLE: Oven temperature 250 ±2°C (482 ±3.6°F)
Dielectric test, 2500 volts (rms) 60Hz
DIELECTRIC TEST AFTER IMMERSION: 2500 volts (rms), 60Hz
ABRASION RESISTANCE AFTER IMMERSION: Qualification test only
HUMIDITY RESISTANCE: 5000 megohms for 1000 ft, min insulation resistance after humidity exposure

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ADDITIONAL REQUIREMENTS (Continued)

SURFACE RESISTANCE:
Sizes 22 through 12: 500 megohm-inches (min), initial and final readings
Sizes 10 through 00: No requirement
SMOKE: 250°C (482°F)
ACID RESISTANCE: Required
Dielectric test, 2500 volts (rms), 60Hz
COLOR: In accordance with MIL-STD-104, Class 1; white preferred

COLOR STRIPING OR BANDING DURABILITY: 250 cycles (500 strokes) (min), 500 grams weight
IDENTIFICATION DURABILITY: 125 cycles (250 strokes) (min); 500 grams weight
WIRE LENGTH REQUIREMENTS: Schedule A
SUPERSESSION DATA: The wire of this specification sheet, by part number, replaces and supersedes the wire of MS90294(AS) (canceled) in accordance with Table III.

TABLE III. SUPERSESSION BY PART NUMBER

Part number MS90294(AS)	Part Number MIL-W-22759/4
MS90294-22-*	M22759/4-22-*
MS90294-20-*	M22759/4-20-*
MS90294-18-*	M22759/4-18-*
MS90294-16-*	M22759/4-16-*
MS90294-14-*	M22759/4-14-*
MS90294-12-*	M22759/4-12-*
MS90294-10-*	M22759/4-10-*
MS90294-8-*	M22759/4-8-*
MS90294-6-*	M22759/4-6-*
MS90294-4-*	M22759/4-4-*
MS90294-2-*	M22759/4-2-*
MS90294-1-*	M22759/4-1-*
MS90294-01-*	M22759/4-01-*
MS90294-02-*	M22759/4-02-*

Custodians:
Navy - AS
Army - EL
Air Force - 99

Review activities:
Navy - EC, OS
Army - ME, MI, MU
Air Force -
DSA - IS
NSA

Preparing activity:
Navy - AS
(Project No. 6145-0701-4)

User activities:
Navy - MC
Army - AT, AV
Air Force - 11

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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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